

## **SQL assignment:**

### **Aggregate Functions**

Question: Find the average salary of employees in the employees table.

Question: Count the number of employees in each department.

Question: Retrieve employee names and their corresponding department names.

Question: Retrieve the names of employees who have a salary higher than the average salary.

### **Complex JOINS**

Question: Retrieve the names of employees and their managers (use self-join).

Question: Retrieve the salaries of employees along with the average salary of their department.

### **CTEs (Common Table Expressions)**

Question: Use a CTE to find the second highest salary in the employees table.

### **Recursive Queries**

Question: Using a recursive CTE, retrieve the hierarchy of employees in the organization.

### **Advanced Aggregation**

Question: Find the top three departments with the highest total salaries.

### **HardComplex JOINS**

Question: Retrieve the names of employees and their managers (use self-join).

Question: Retrieve the salaries of employees along with the average salary of their department.

### **Recursive Queries**

Question: Using a recursive CTE, retrieve the hierarchy of employees in the organization.

### **Advanced Aggregation**

Question: Find the top three departments with the highest total salaries.

1. Create an ER diagram for a university database that includes entities such as Students, Courses, Instructors, and Departments. Define the relationships among these entities. Hint: Identify primary and foreign keys, and consider relationships like "enrolls," "teaches," and "belongs to."
2. Convert the ER diagram you created into a relational schema. Clearly define tables, primary keys, and foreign keys. Hint: Break down the entities into tables and show how they relate through keys.
3. Given a table with the following structure, normalize it to 3NF:  
Employee (EmployeeID, EmployeeName, DepartmentID, DepartmentName, ProjectID, ProjectName, HoursWorked) Hint: Start by removing partial and transitive dependencies.
4. Design a simple transaction in SQL to transfer money between two accounts. Ensure that your transaction adheres to the ACID properties. Hint: Use BEGIN TRANSACTION, COMMIT, and ROLLBACK.
5. Discuss different types of indexes and their use cases.

**MAIN QUESTION:**

Design a small database for a library system, including entities such as Books, Members, Loans, and Authors. Provide the ER diagram, relational schema, and sample SQL queries to manage the system. Hint: Include queries for checking out books, returning books, and querying overdue loans.